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**Amirkabir University of Technology**  
**Course: Autonomous Aerial Mobile Robots**  
**3D Vision Project**

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## Part I) Camera Calibration

Every practical 3D vision pipeline starts with sensor calibration. As discussed in class, reliable geometric reconstruction depends on estimating these parameters accurately.

- 1) Use [this notebook](#) as a starting point to calibrate your mobile phone camera. Document your calibration procedure and report the resulting parameters. Calibration patterns may be generated from [this website](#).

## Part II) Stereo Vision

Stereo vision offers an affordable route to metric 3D reconstruction. In this part, you will estimate scene depth using the [Kagaru Airborne dataset](#). Download the archive [100831\\_155323\\_MultiCamera0\\_subset\\_db.rar](#), which contains 100 frames from the dataset. The intrinsic and stereo parameters for the two cameras are available on the same webpage.

- 2) Using OpenCV, compute the depth map for each of the 100 frames and include the resulting depth sequence as a video attachment to your report.
- 3) Back-project the depth maps into point clouds using the known camera intrinsic parameters.

## Part III) Single-view Geometry (Bonus)

Although a single image does not directly reveal full 3D structure, useful geometric quantities can still be recovered when prior information is available. Suppose the true height of one object in the scene below, such as a traffic light, is known.



- 4) Assuming the camera intrinsic parameters are given, derive how the camera height above the ground plane can be estimated.

**Hint.** Estimating the camera tilt from at least one vanishing point is a natural place to begin; see the appendix.

## Appendix: Vanishing Points

In Euclidean geometry, parallel lines never meet. In projective geometry, however, they intersect at a point at infinity, which appears in the image plane as a *vanishing point*. In the example below, the vanishing point can be identified by extending at least two parallel field markings. Any other line passing through that point belongs to the same family of parallel lines. This idea is used in practical tasks such as offside analysis in sports footage.

Vanishing points play a central role in 3D computer vision. They are especially useful for camera calibration and for recovering measurements from a single image. For this assignment, if the vanishing point associated with the ground plane is available, the camera pitch angle may be computed as

$$\theta_{\text{pitch}} = \tan^{-1} \left( \frac{v_y - c_y}{f} \right).$$

Here,  $v_y$  and  $c_y$  denote the vertical pixel coordinates of the vanishing point and the image center, respectively, while  $f$  is the focal length. Keep in mind that the vanishing point does not have to lie inside the image boundaries.

